

AE516 - Final Project - Report

Vishwa Mehta

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Introduction:

Problem Description:

The aim of this project is to design a 12-ply laminate using the classical lamination theory to meet the flexural modulus of 93GPa , using IM7-8552 Pre-Preged Carbon Fiber plies. The selection of the laminate is to be based on the material being wasted for the manufacturing, tensile stiffness and the maximum tensile strength of the laminate based on the Tsai-Hill Failure Criteria. Hence, after finding the ply orientations that satisfy the requirement of the flexural modulus, the laminates are sorted based on their calculated waste material (0° and 90° being 0 wastage) and their tensile strengths as calculated by the algorithm that keeps increasing the longitudinal load on the laminate and removes the plies progressively as they fail. Hence finding the highest tensile load that the laminate can withstand before all the plies reach failure based on the Tsai-Hill Failure Criteria.

Material Properties:

For manufacturing the laminate, Hexcel IM7-8552 (Unidirectional) is used as the raw material, that is a Pre-Preged Carbon Fiber with properties as mentioned below: (Source: Hexcel 8552 IM7 Unidirectional Prepreg Qualification Material Property Data Report)

Property	Value
Longitudinal Tensile Modulus (E_1)	158.5105 GPa (22.99 Msi)
Transverse Tensile Modulus (E_2)	8.9632 GPa (1.3 Msi)
Shear Modulus: G_{12}	4.6884 GPa (0.68 Msi)
Poisson's Ratio (ν_{12})	0.316
Longitudinal Tensile Strength (S_L)	2500 MPa
Transverse Tensile Strength (S_T)	64.05 MPa
Shear Strength (S_{LT})	91.14 MPa

Design Methodology:

Classic Lamination Theory:

The Classic Lamination Theory allows us to find the resultant properties of any laminate if we know the individual properties of its constituent plies. The physical properties of the lamina are utilized to find the $[\mathbf{Q}]$ matrix, and is used to find the $[\bar{\mathbf{Q}}]$ matrix using the transformation matrix ($[\mathbf{T}]$) for the orientation of that lamina. The $[\bar{\mathbf{Q}}]$ matrix of each lamina is then used according to their arrangement in the laminate to find the $[\mathbf{A}]$, $[\mathbf{B}]$ and $[\mathbf{D}]$ matrices of the lamina.

These matrices are calculated using:

$$A_{ij} = \sum_{k=1}^N (\overline{Q}_{ij})_k (z_k - z_{k-1}) \quad (1)$$

$$B_{ij} = \frac{1}{2} \sum_{k=1}^N (\overline{Q}_{ij})_k (z_k^2 - z_{k-1}^2) \quad (2)$$

$$D_{ij} = \frac{1}{3} \sum_{k=1}^N (\overline{Q}_{ij})_k (z_k^3 - z_{k-1}^3) \quad (3)$$

To find the Flexural Modulus,

$$D^* = D - B \cdot A^{-1} \cdot B \quad (4)$$

$$D' = (D^*)^{-1} \quad (5)$$

$$E_{fx} = \frac{12}{t^3 D'_{11}} \quad (6)$$

Similarly, to find the Tensile Modulus,

$$A^* = A^{-1} \quad (7)$$

$$C^* = B \cdot A^{-1} \quad (8)$$

$$A' = A^* - B \cdot (D^*)^{-1} \cdot C^* \quad (9)$$

$$E_x = \frac{1}{t A'_{11}} \quad (10)$$

These values are found for all the laminates in the selection algorithm, and sorted accordingly.

Selection Algorithm:

The selection algorithm for the desired laminate first sorts out all the possible ply-orientations of the laminate that have the required value of the flexural modulus (around a pre-set tolerance value, found using the CLT). The algorithm then calculates the Tensile Strengths of all the eligible laminates and sorts them in the order of highest to lowest tensile strength value. The laminates with the highest possible Tensile Strengths are then evaluated based on the theoretical material wastage occurring during their manufacturing and their respective Tensile Stiffness values (found using the CLT), and are sorted based on the material wastage. The final list of favourable laminates are then manually evaluated based on the trade-offs between the material wastage and the Tensile Stiffness values. The algorithms used for determining the Tensile Strengths and Material Wastage are as follows:

Tensile Strength Evaluation:

For a given lamina layup, the effective Longitudinal Tensile Strength is calculated for each constituent lamina using the theoretical value provided by the Tsai-Hill criterion for off-axis tensile test:

$$\sigma_x = \left(\frac{1}{\frac{\cos^4 \theta}{S_L^2} + \left(\frac{1}{S_{LT}^2} + \frac{1}{S_L^2} \right) \sin^2 \theta \cos^2 \theta + \frac{\sin^4 \theta}{S_T^2}} \right)^{\frac{1}{2}} \quad (11)$$

where θ is the orientation angle of the respective ply. Hence, after finding the effective tensile strengths of each ply in the longitudinal direction, the algorithm simulates a progressive tensile load on the laminate where the load is divided equally among all the 'active' plies (a ply is active if its tensile strength is higher than the load being applied on it, hence when it has not failed yet). As the load on a ply exceeds its tensile strength value, it is marked as 'inactive' (or failed), and the load is redistributed among the active plies. This process is repeated and the load is increased till all the constituent plies achieve failure, and the final load is said to be the Longitudinal Tensile Strength of the laminate.

Material Wastage Evaluation:

The total material wastage for manufacturing a laminate was calculated by accounting for the material wasted for each constituent lamina assuming it was to be cut from a square piece of pre-preg fabric with the desired ply orientation angle as shown in the figure below:

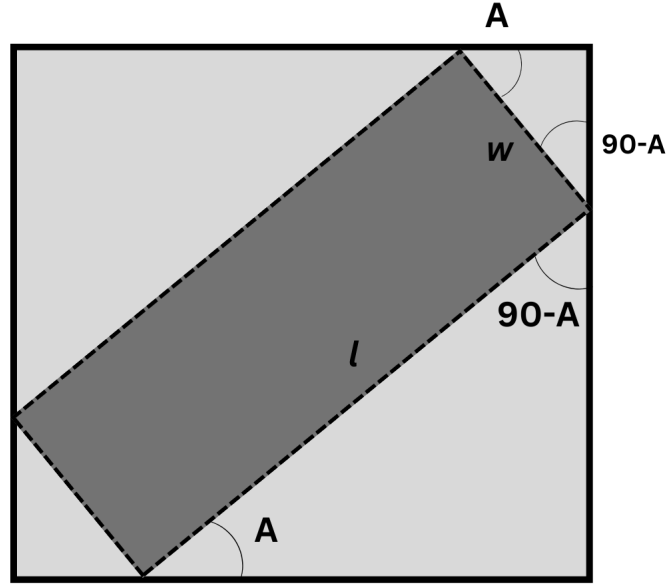


Figure 1: Material Wastage upon cutting a lamina (with A° orientation)

The waste material for a lamina with A° ply orientation is calculated using trigonometry and is found to be:

$$Waste_A = 2\left(\frac{1}{2}l^2 \cos A \sin A + \frac{1}{2}w^2 \cos A \sin A\right) \quad (12)$$

$$Waste_A = (l^2 + w^2) \cos A \sin A \quad (13)$$

The sum of material wasted for each lamina is hence the material wastage value for manufacturing the laminate.

Initializing the Algorithm:

To find the best possible laminate while keeping the computational load in the feasible region, the algorithm is initialized to have possible orientation values ranging from -90° to $+90^\circ$ with 15° variations, and the acceptable tolerance is set at 0.1% .

Design provided by the algorithm:

At a tolerance of 0.1% , the final design provided by the algorithm was:

Orientation: $[75^\circ, 0^\circ, 0^\circ, 0^\circ, 30^\circ, 0^\circ, 0^\circ, 30^\circ, 0^\circ, 0^\circ, 0^\circ, 75^\circ]$

$E_{fx} : 93.3036GPa$

$E_x : 116.7481GPa$

$S_L : 1667.2727MPa$

$Wastage : 67.471Inch^2$

Final Design:

The final design was chosen by conducting a manual trade-off evaluation between the designs provided by all the group members. The design in competition with the aforementioned design was provided by fellow graduate student Sinaendhran. The design chosen was:

Orientation: $[90^\circ, 0^\circ, 0^\circ, 0^\circ, 30^\circ, 0^\circ, 0^\circ, 30^\circ, 0^\circ, 0^\circ, 0^\circ, 90^\circ]$

$E_{fx} : 93.3429GPa$

$E_x : 115.890GPa$

$S_L : 1667.2727MPa$

$Wastage : 38.9711Inch^2$

The main trade-off that became the deciding factor was the material wastage between the 2 designs. Hence, the team went forward to manufacture this design instead, as the remaining criteria were almost identical in both the designs, at the risk of having a slightly less accurate value of the flexural modulus than what was desired.

The above mentioned algorithm did not initially converge to this design due to the strict tolerance of 0.1% being used, which was set to minimize the error between the manufactured laminate and the theoretically obtained laminate. The algorithm, when tested with a relatively relaxed tolerance of 0.5% , converged to the same final layup.

Manufacturing Process:

Cutting of Laminae and Layup Preparation:

To manufacture the finalised design,

Cutting of Laminae:

A sheet of unidirectional IM7-8552 prepreg is used to cut out the required plies (measuring 3 inches x 6 inches). 8 plies of 0° orientation, 2 plies of 90° orientation and 2 plies of 30° orientation are cut out. The laminae are then arranged in the required stacking order.

Processes done before curing:

The metal plate upon which the curing is to be done inside the autoclave is cleaned with ethanol. A square is made on the plate using vacuum tape with each side measuring 12 inches. A mold release solution is then applied in the interior area. The metal plate being used to ensure uniform pressure is also applied mold release on. 2 plies of the peel-ply release fabric are cut that are slightly bigger than the size of the pre-preg plies. A single layer of the breather-fabric is also cut that is big enough to cover the interior of the vacuum tapes. The fabrics are then arranged in the order (starting from bottom): peel-ply, lamina stack, peel-ply, and then this stack is put under the metal plate for equal pressure. The metal plate is surrounded by resin dams to not let the resin ooze out of the layup during the curing process. A pipe is then attached on top of 1 edge of the vacuum tape (that will connect to the vacuum during curing). The entire stack is then covered by the breather-fabric such that the pipe is also covered by it. The entire assembly is then covered by a vacuum bag and a seal is ensured on all edges of the vacuum tapes. Vacuum testing is done for the assembly and then it is ready for the curing process.

Curing Process:

The assembly is then put inside the autoclave while being connected to a constant vacuum to ensure all bubbles from within the pre-preg exit when the temperature rises. The autoclave is set at a constant temperature of 180°C for 2-3 hours. In addition to this, a constant external pressure of 3-4 bars is also applied to ensure uniform dimensions of the resultant laminate.

Post Processing of Test Samples:

Cutting and Sanding:



Figure 2: Manufactured Laminate

The Laminate hence formed is first cut using a diamond-saw if necessary (to remove the uneven bulged part on the edges formed due to uneven pressure). The remaining laminate is then cut using a Water-Jet, to cut 5 samples for 4-point bend testing. While cutting, the Water-Jet cutting path is determined by measuring the average thickness of the laminate to ensure the 1:40 ratio of thickness to span required for the test. A minimum of 20% extra length is also ensured in the length of the samples to abide by the requirements of ASTM - D7264 (to ensure that slipping does not take place when the sample is bent on the pins of the Instron. The cut samples are then sanded to remove the tabs (that are set in the path of the water-jet to keep the cut samples attached from the laminate after being cut) to hence obtain the test-ready samples.

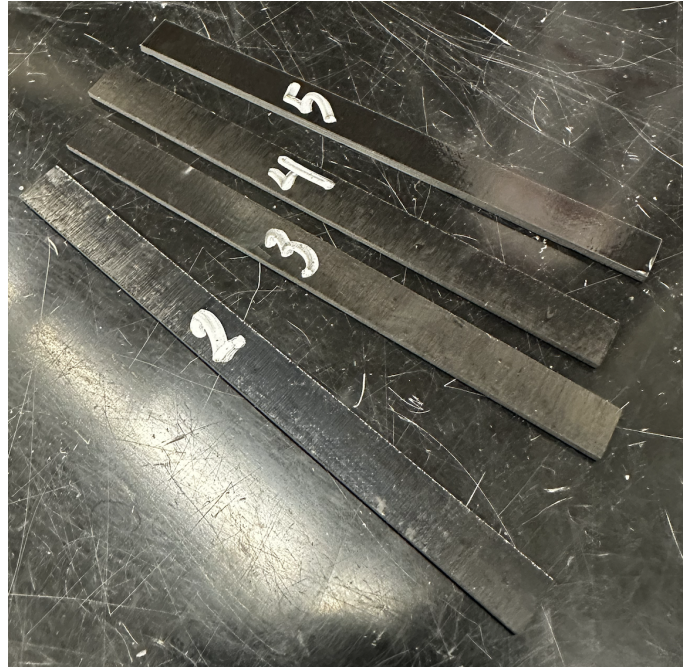


Figure 3: Post-Processed Test Specimens

Test Sample Specifications:

The resultant samples are then measured (using Screw-Gauge and Vernier Calipers) for their final dimensions to be used for calculation of the Flexural Modulus post-testing. The recorded mean values are:

Sample Number	Thickness (in mm)	Width (in mm)
1	2.007	10.07
2	1.983	10.077
3	2.07	10.05
4	2.023	10.007
5	2.12	10.03

Hence, the mean thickness (for calculation of Span) is found to be: $2.047mm$

Experimentation:

ASTM D7264:

ASTM - D7264 is the Standard Test Method for Flexural Properties of Polymer Matrix Composite Materials. There are 2 testing methods, Procedure A (for 3-Point Bend Test) and Procedure B (for 4-Point Bend Test). For this project, Procedure B is undertaken as the testing method.

Procedure B is defined as a four-point loading system utilizing two load points equally spaced from their adjacent support points, with a distance between load points of one-half of the support span.

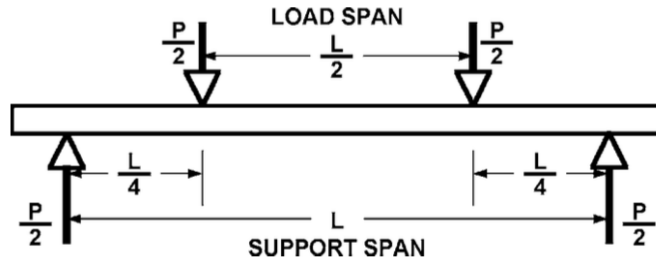


Figure 4: 4-Point Bending Loading Diagram (Source: ASTM - D7264)

As depicted in the diagram, the test sample is loaded at quarter-span positions by the loading pins, and held by the bottom pins at a span length determined by the sample's thickness. Among the possible thickness to span ratios, 1 : 40 is chosen for this project.

The sample is progressively loaded until it fails or till the maximum possible deflection (set in the Intron) is achieved. The corresponding load - deflection values are then utilized to find the Flexural Modulus of the laminate samples.

Testing Specifications:

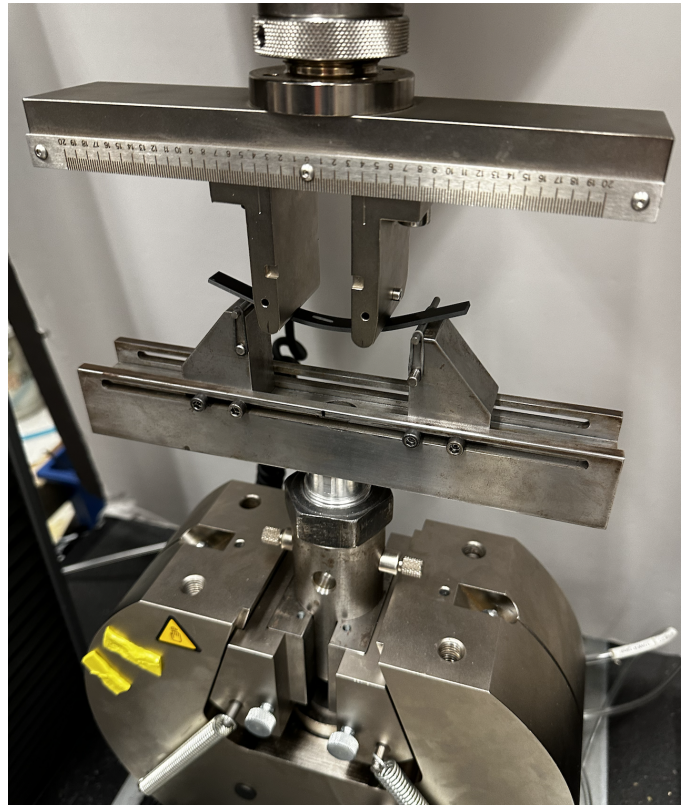


Figure 5: Testing a Specimen using the 4-Point Bend Method

For our testing, based on the average thickness of the test samples, and the 1:40 thickness to span ratio, the **Span Length** is set at 81.86mm . And the corresponding **Half-Span Length** is set at 40.93mm for the testing.

Other machine specific properties are set to be:

Rate of Deflection: $1mm/min$

Maximum Deflection: $5mm$

Post-Processing Test Data:

The test data obtained for each test sample includes the time stamp of the test with the quarter-span (loading pin) deflection and corresponding force applied. This data is used to obtain the Flexural Modulus using the formulae given below:

$$E_{fx} = \frac{Pl^3}{8wbh^3} \quad (14)$$

where P is the load applied, l is the span length, w is the quarter-span deflection, b is the width of the specimen, and h is the thickness of the specimen. (Source: Composite Materials: Testing and Design, Chapter 2, Equation 31)

The flexural modulus given by this equation is used for all data-points for the given span, and the average of all values is taken to obtain the flexural modulus of that specimen. The same is repeated for all specimens to obtain the flexural modulus of the manufactured laminate.

Results and Discussion:

The resultant values of flexural modulus for each test specimen hence obtained experimentally is as follows:

Specimen Number	E_{fx} (in GPa)
1	93.899
2	100.483
3	98.777
4	97.669
5	98.158

Mean Flexural Modulus: $97.797GPa$

Percentage Error between the obtained and desired values:

$$\%Error = \frac{97.797 - 93}{93} \% = 4.90\% \quad (15)$$

The error lies within the range of acceptable values between computational estimations and physical values obtained after manufacturing and testing. (the causes of error can include theoretical approximations in the CLT, manufacturing defects and experimental error during testing)

Conclusion:

This project encapsulated the entire process starting from designing a composite laminate using algorithmic models following the Classical Lamination Theory, sorting the favourable laminates based on trade-offs in the values of Material Waste, Tensile Stiffness and Tensile Strengths, hence coming up with the highest possible values of Tensile Strength and Stiffness while theoretically minimizing

the material wastage during manufacturing. The laminate was then manufactured (and tested) in the **Aerospace Materials Laboratory at the University of Michigan**. The flexural modulus of the laminate was then tested using the 4-point bend method given in ASTM - D7264 and the experimental values were compared with the expected value which provided a satisfactory result.

References:

- (ASTM D7264/D7264M-07)
- Whitney, J. M., Browning, C. E., and Mair, A., “Analysis of the Flexure Test for Laminated Composite Materials,” Composite Materials: Testing and Design (Third Conference), ASTM STP 546, 1974, pp. 30-45
- Hexcel 8552 IM7 Unidirectional Prepreg 190 GSM & 35%RC Qualification ..., www.wichita.edu/industry_and_defense/NIAR/Documents/Qual-CAM-RP-2009-015-Rev-B-Hexcel-8552-IM7-MPDR-04.16.19.pdf. Accessed 13 Dec. 2023.